

$$\text{Soru 1: } \cancel{3-5} \cdot 3 - 5 \cdot [2 - (1-8) \cdot 4 - 3 \cdot (-5)] - 2 - 5 \cdot (3-4) = ?$$

$$3 - 5 \cdot [2 - (-7) \cdot 4 + 15] - 2 - 5 \cdot (-1)$$

$$3 - 5[2 + 28 + 15] - 2 + 5 = 3 - 5 \cdot 45 - 2 + 5$$

$$3 - 225 + 3$$

$$= -219$$

$$\text{Soru 2: } (-3)^2 - [(-30) : 3 + 2 \cdot (-5) - 8 \cdot (2-5)] \cdot 2 - 9 = ?$$

$$9 - [-10 - 10 - 8 \cdot (-3)] \cdot 2 - 9$$

$$9 - (-20 + 24) \cdot 2 - 9$$

$$\cancel{9-8-9} = -8$$

$$\text{Soru 3: } x - \{1 - [2x - 3 \cdot (x+1)]\} + 4 \cdot (2-2x) = ?$$

$$x \{1 - [2x - 3x - 3]\} + 8 - 8x$$

$$x - \{x + 4\} + 8 - 8x$$

$$\cancel{x-x} - 4 + 8 - 8x = -8x + 4$$

$$\text{Soru 4: } -3[(x+2y) - 3 - 2 \cdot (x+y)] - [(3x+1) - (x-8)] = ?$$

$$-3 \cdot (x+2y - 3 - 2x - 2y) - (3x+1 - x+8)$$

$$-3(-x-3) - (2x+9)$$

$$3x+9-2x-9 = x$$

$$\text{Soru 5: } \frac{5}{6} + \frac{4}{5} + \frac{2}{3} = ?$$

$$\frac{5}{6} + \frac{4}{5} + \frac{2}{3} = \frac{25}{30} + \frac{24}{30} + \frac{20}{30} = \frac{69}{30} = \frac{23}{10} = 2 \frac{3}{10}$$

$$\text{Soru 6: } 5 \frac{1}{2} - \frac{3}{5} = ?$$

$$5 \frac{1}{2} - \frac{3}{5} = \frac{11}{2} - \frac{3}{5} = \frac{55}{10} - \frac{6}{10} = \frac{49}{10} = 4 \frac{9}{10}$$

$$\text{Soru 7} = 2 \frac{3}{4} \cdot \left(-3 \frac{5}{6}\right) \cdot \frac{24}{11} = ?$$

$$2 \frac{3}{4} \cdot \left(-3 \frac{5}{6}\right) \cdot \frac{24}{11} = \frac{11}{4} \cdot \left(-\frac{23}{6}\right) \cdot \frac{24}{11} = -23$$

$$\text{Soru 8} : \frac{3}{4} : \frac{2}{3} = ?$$

$$\frac{3}{4} : \frac{2}{3} = \frac{3}{4} \cdot \frac{3}{2} = \frac{9}{8}$$

$$\text{Soru 9} = 2 \frac{1}{5} : 1 \frac{1}{3} = ?$$

$$2 \frac{1}{5} : 1 \frac{1}{3} = \frac{11}{5} : \frac{4}{3} = \frac{11}{5} \cdot \frac{3}{4} = \frac{33}{20}$$

$$\text{Soru 10} \Rightarrow \left(2 \frac{1}{3} + 1 \frac{1}{4}\right) - \left(1 \frac{1}{2} - 1 \frac{1}{4}\right) = ?$$

$$\left(\frac{7}{3} + \frac{5}{4}\right) - \left(\frac{3}{2} - \frac{5}{4}\right) = \frac{28+15}{12} - \frac{6-5}{4} = \frac{43}{12} - \frac{1}{4} = \frac{43-3}{12} = \frac{40}{12} = \frac{10}{3}$$

$$\text{Soru 11} \Rightarrow \left(1 \frac{2}{3}\right)^{-1} + \left(\frac{2}{7}\right)^{-1} = ?$$

$$\left(\frac{5}{3}\right)^{-1} + \frac{7}{2} = \frac{3}{5} + \frac{7}{2} = \frac{6+35}{10} = \frac{41}{10} = 4 \frac{1}{10}$$

$$\text{Soru 12} \Rightarrow \left(\frac{1}{5} : \frac{1}{3}\right) + \left(\frac{3}{4} - \frac{1}{8}\right) = ?$$

$$\left(\frac{1}{5} \cdot \frac{3}{1}\right) + \left(\frac{3}{4} - \frac{1}{8}\right) = \frac{3}{5} + \frac{5}{8} = \frac{24+25}{40} = \frac{49}{40} = 1 \frac{9}{40}$$

$$\text{Soru 13} \Rightarrow \left(\frac{1 - 1\frac{1}{3}}{1 + 1\frac{1}{3}} \right)^2 \cdot \left(\frac{7}{2} \right)^2 = ?$$

$$\left(\frac{-\frac{1}{3}}{\frac{4}{3}} \right)^2 \cdot \frac{49}{4} = \left(-\frac{1}{3} \cdot \frac{3}{4} \right)^2 \cdot \frac{49}{4} = \left(-\frac{1}{4} \right)^2 \cdot \frac{49}{4}$$

$$= \frac{1}{16} \cdot \frac{49}{4} = \left(\frac{1}{4} \right)$$

Soru 14 $\Rightarrow a^{-1} = \frac{1}{5}$ ve $b^{-1} = \frac{1}{4}$ olduğuna göre $\frac{1}{a} - \frac{1}{b}$ ifadesi neye eşittir?

$$\begin{array}{l} a^{-1} = \frac{1}{5} \text{ ise } a = 5 \\ b^{-1} = \frac{1}{4} \text{ ise } b = 4 \end{array} > \frac{1}{5} - \frac{1}{4} = \frac{4}{20} - \frac{5}{20} = \left(-\frac{1}{20} \right)$$

$$\text{Soru 15} \Rightarrow 1 + \frac{1}{1 + \frac{1}{1 + \frac{1}{1 + \frac{1}{3}}}} = 1 + \frac{1}{1 + \frac{1}{\frac{4}{3}}} = 1 + \frac{1}{1 + \frac{3}{4}}$$

$$= 1 + \frac{1}{\frac{7}{4}} = 1 + \frac{4}{7} = \left(\frac{11}{7} \right)$$

$$\text{Soru 16} \Rightarrow 1 + \frac{\frac{1}{2} + \frac{2}{3}}{1 + \frac{1}{1 + \frac{1}{2}}} = ? \quad 1 + \frac{\frac{1}{2} + \frac{8}{3}}{1 + \frac{1}{\frac{3}{2}}} = 1 + \frac{\frac{19}{6}}{1 + \frac{2}{3}}$$

$$= 1 + \frac{\frac{19}{6}}{\frac{5}{3}} = 1 + \frac{19}{6} \cdot \frac{3}{5} = 1 + \frac{19}{10} = \left(\frac{29}{10} \right)$$

Soru 17 $\Rightarrow \frac{1}{4}$ kesriyi ondalik desir olarak yaziniz.

$$\frac{1}{4} = \frac{25}{100} = 0,25$$

Soru 18 $\Rightarrow 12,37 + 2,084 = ?$

$$\begin{array}{r} 12,37 \\ + 2,084 \\ \hline 14,454 \end{array}$$

Soru 19 $\Rightarrow 8,04 - 3,325 = ?$

$$\begin{array}{r} 8,040 \\ - 3,325 \\ \hline 4,715 \end{array}$$

Soru 20 $\Rightarrow 36,5 \times 0,24 = ?$

$$\begin{array}{r} 36,5 \\ \times 0,24 \\ \hline 1460 \\ 730 \\ 000 \\ \hline 8,76 \end{array}$$

Soru 21 $\Rightarrow \frac{0,02}{0,004} + \frac{0,03}{0,005} + \frac{0,04}{0,08} = ?$

$$\frac{0,02}{0,004} + \frac{0,03}{0,005} + \frac{0,04}{0,08} = \frac{20}{4} + \frac{30}{5} + \frac{400}{8} = 5 + 6 + 50 = 61$$